

BRC WRF State Playbook

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Short print-oriented explanation for John/JRL and Michael. It describes where the current `brc-wrf` fork fits, what is proven, and what should happen next.

One-Sentence State

We have a BRC-local review layer around WRF 4.8.0, and one NAM-only Jan-2013 Basin case is proven through `brc-tools` input staging, WPS, `real.exe`, `wrf.exe`, archive checks, and no-run quicklooks.

Mental Model

LAYER	OWNS	DOES NOT OWN
<code>brc-tools</code>	Downloading/staging WRF-ready GRIB files, manifests, contracts, input quicklooks.	WPS, <code>real.exe</code> , <code>wrf.exe</code> , Slurm run wrappers.
<code>brc-wrf</code>	WRF source, WPS/WRF consumption docs, case manifests, validators, Slurm rendering, WRF-output quicklooks.	NWP downloader logic.
<code>brc-knowledge</code>	Canonical CHPC hardware, storage, scheduler, and validated example Slurm scripts.	Repo-local code or case manifests.

The AI tooling is useful because it keeps these contracts visible and checks cheap facts before expensive runs. It is also another layer to audit, so every AI-assisted step should leave a small, readable breadcrumb: command, evidence, owner repo, and stop point.

Who Can Understand This Today?

AUDIENCE	CURRENT READINESS	WHAT STILL FEELS HARD
John/JRL	High enough to audit and steer science decisions.	The GEFS+NAM two-stream path still needs WPS/ <code>real.exe</code> proof before it is science-ready.
Michael/new developer	Medium if starting from the reading packet below.	WRF requires both software-install knowledge and meteorological forcing knowledge; the repo split must be read first.
Future AI agent	Medium-high for no-run review tasks.	It must not confuse old NAM-only proof artifacts with unproven GEFS+NAM workflow.

Where We Are

AREA	STATUS
Fork orientation	Usable. <code>README.md</code> , <code>AGENTS.md</code> , and <code>doc/BRC_FORK_GUIDE.md</code> explain the local layer.
CHPC posture	Usable. Canonical CHPC facts live in <code>brc-knowledge</code> ; this repo points there.
First case	Proven NAM-only path for Jan 31-Feb 2 2013, d01/d02 Basin nest.
Case review	Usable. <code>brc-cases/wrf_case.py</code> validates metadata and renders Slurm text only.
Visual QA	Usable. <code>brc-cases/wrf_quicklook.py</code> renders five no-run PNGs from existing proof artifacts.
Slurm profile	Aligned to max owned-node profile: <code>lawson-np</code> , <code>notch392</code> , 1 node, 56 tasks, <code>900G</code> , <code>srun --mpi=pmi2</code> .
GEFS+NAM	Not proven. Treat as a design/proof task, not a working production method.

Where We Should Go Next

ORDER	NEXT MOVE	STOP POINT
1	Review the quicklook PNGs with meteorological eyes.	Decide whether the NAM-only proof looks physically plausible enough to keep as baseline.
2	Commit this docs/validator/quicklook checkpoint.	Branch has a clean, reviewable milestone.

ORDER	NEXT MOVE	STOP POINT
3	Push or otherwise account for the current local commit stack.	Remote and local state are not ambiguous.
4	Decide whether GEFS+NAM is still needed for the next science question.	If yes, draft the two-stream WPS proof; if no, improve NAM-only repeatability.
5	For any real run, render Slurm and inspect it against <code>brc-knowledge</code> before <code>sbatch</code> .	Human approval only after the rendered script matches current CHPC truth.

Reading Packet

Read these in order for a milestone review:

1. `brc-docs/BRC-WRF-MICHAEL-PRACTICAL-PACKET.md` for a pair-programming walkthrough.
2. `brc-docs/BRC-WRF-FORK-HIGHLIGHTS.md`
3. `brc-docs/BRC-WRF-FIRST-CASE.md`
4. `brc-cases/README.md`
5. `brc-docs/BRC-WRF-USAGE.md`
6. `../brc-tools/docs/WRF-STAGING-STATE-PLAYBOOK.md`
7. `brc-docs/BRC-TOOLS-LINK-HANDOFF.md` if opening a `brc-tools` session
8. `../brc-knowledge/scholarium/reference-base/resources/chpc-team-resource-inventory.md` sections 1-3 and Q1
9. `../brc-knowledge/scholarium/reference-base/resources/wrf-on-chpc-quickstart.md` sections 2, 3, and 8

For Michael, start with items 1, 3, 4, and 6 before the full CHPC resource inventory. For John, start with items 2, 3, 6, 8, and 9; add item 7 when the next task is in `brc-tools`.

Maximum Owned-Node WRF Profile

Use this when the goal is a high-powered, non-preemptible single WRF run on the team-reserved node:

SLURM SETTING	VALUE	SOURCE
account	<code>lawson-np</code>	<code>brc-knowledge</code> resource inventory
partition	<code>lawson-np</code>	<code>brc-knowledge</code> resource inventory
node	<code>notch392</code>	preferred WRF workhorse
nodes	<code>1</code>	avoids asymmetric two-node WRF costs
tasks	<code>56</code>	all Slurm cores on <code>notch392</code>
memory	<code>900G</code>	reserves most of the ~996 GiB node
launcher	<code>srslurm --mpi=pmi2</code>	validated Intel MPI fix

SLURM SETTING	VALUE	SOURCE
preemptible?	no	owned partition

Do not move to preemptible Granite or multi-node Notchpeak until the case needs more than one notch392 can provide.